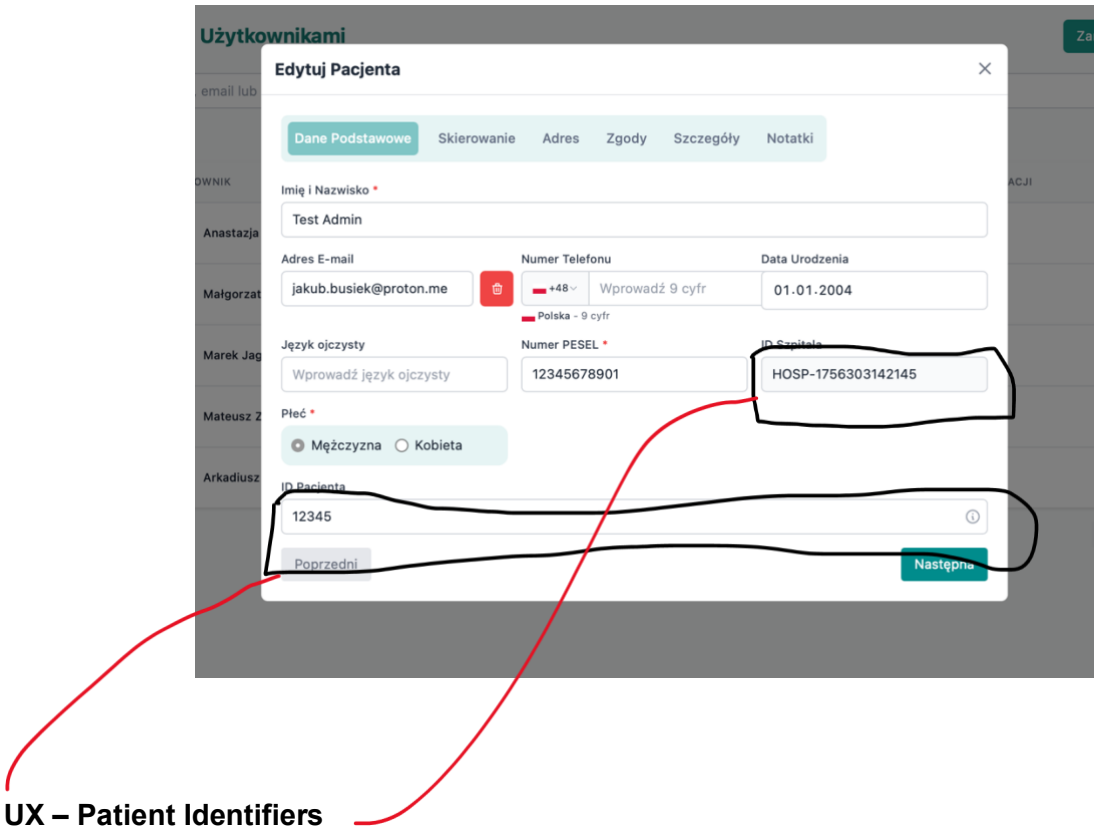
SPECIFICATION – PATIENT / VISIT / REGISTRATION LOGIC (FOR IMPLEMENTATION)

1. CORE SYSTEM PRINCIPLE (OVERRIDING RULE)

- **Patient ≠ Visit**
- A **Visit (VISIT_ID)** may exist independently, without a patient.
- A **Patient (PATIENT_ID)** is created **ONLY** after:
  - entering a valid **PESEL**, and
  - clicking **"Complete registration"**.



UX – Patient Identifiers

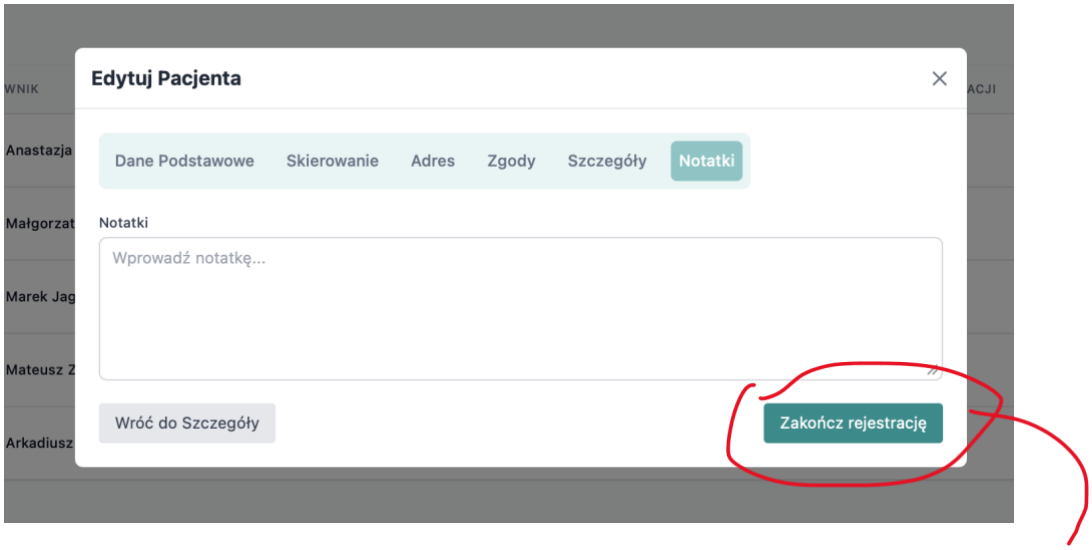
- **Hospital ID is deprecated and must be removed** from the system (UX and logic).
- **Patient ID must never be manually editable or entered by the user.**

UX behavior:

- If **Patient ID does not exist** (visit-only state):
  - display the following message:

    „Identyfikator pacjenta zostanie utworzony po zakończeniu rejestracji pacjenta w systemie."

- If **Patient ID exists**:
  - display the Patient ID as **read-only text** (view only, no editing).



this button "Finish registration" causes the patient's patient ID to be created after entering the PESEL number and a given visit is assigned to his patient ID

2. IDENTIFIERS – WHAT IS CREATED AND WHEN

VISIT_ID

(Already exists in the system and is present in the visit URL / link.)

- **Always created** when a visit is registered:
  - online
  - by reception staff
- Exists **independently of the patient**, because a visit is defined by the appointment itself, not by the patient entity.
- May **permanently exist without a PATIENT_ID**:
  - if the patient does not show up,
  - if PESEL is never entered,
  - if registration is not completed,
  - in such cases the visit remains as a cancelled / no-show VISIT_ID without creating unnecessary patient accounts.

PATIENT_ID

- Created **ONLY when**:
  - a valid **PESEL** is entered, and
  - **"Complete registration"** is clicked.
- **PESEL is the only unique identifier** for a patient
  (phone number is NOT unique and must not be used as a patient key).
- **No PESEL = no patient.**

USERNAME (Patient Account)

- Created **ONLY for patients who have a PATIENT_ID**.
- **NEVER created** for:
  - cancelled or no-show visits,
  - persons who have never physically visited the clinic.
- **USERNAME is generated only upon patient request**, if the patient decides to use the patient portal
  (details described in a later section).

3. VISIT REGISTRATION – SCENARIOS

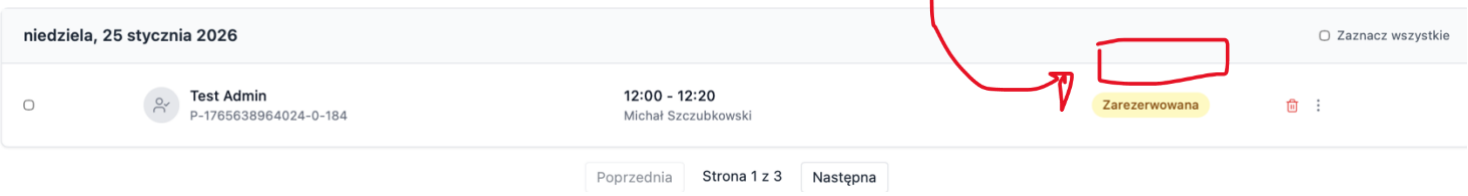
A. ONLINE Registration

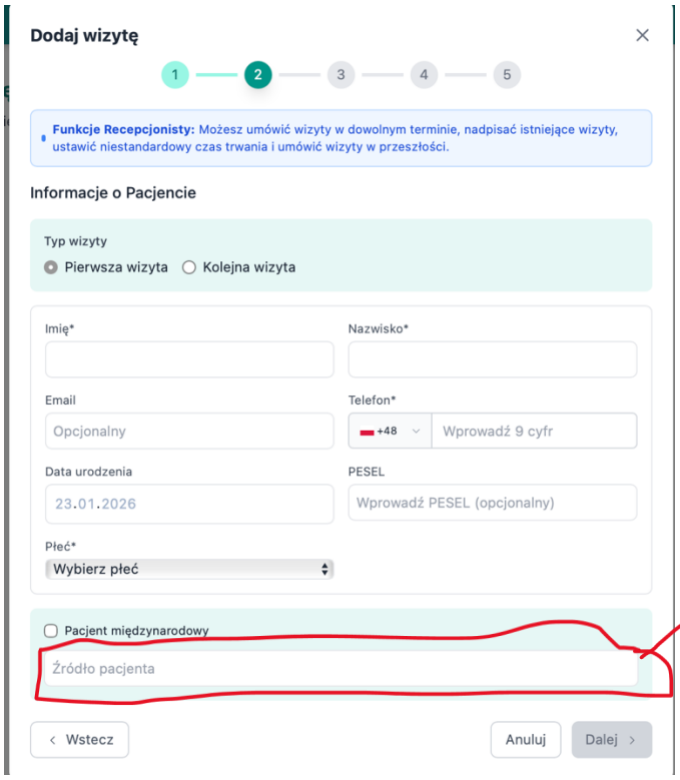
System behavior:

- The system:
  - creates a **VISIT_ID**,
  - stores all required data submitted in the online registration form (as currently implemented),
  - sets: booking_source = ONLINE (which indicates that the visit was registered directly by the patient and allows this to be displayed in the UX.)

- The system **does NOT create**:
  - a **PATIENT_ID**,
  - a **USERNAME** (patient account).

- Next to the visit status, display the label: **„Rejestracja online”**





Rremove this patient source field because it has no significance in the system or use it for something key and important for the functioning of the dashboard

B. Registration by RECEPTION – “Follow-up Visit”

Reception staff:

- selects an existing patient.

System behavior:

- creates a VISIT_ID,
- immediately assigns:
  VISIT_ID → PATIENT_ID
- sets:
  booking_source = RECEPTION

UX:

- no label is displayed (intentional informational silence).

C. Registration by RECEPTION – “First Visit”

Reception staff:

- enters the same basic data as currently required,
- phone number is optional.

System behavior:

- creates a VISIT_ID,
- **does NOT create:**
  - **PATIENT_ID,**
  - **USERNAME.**

Phone number rules:

- **phone number is NOT unique,**
- the same phone number may be associated with **multiple visits** and/or persons (e.g. a parent registering multiple children).

4. UI TABS – RULES (VERY IMPORTANT)

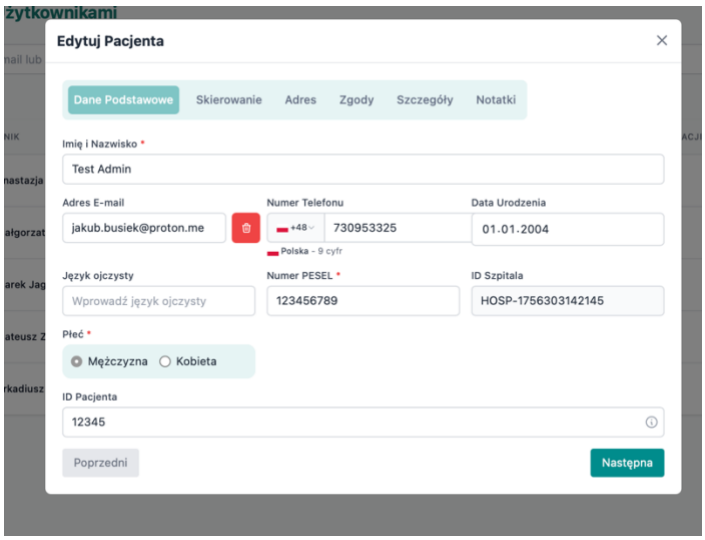
ALWAYS visible (VISIT_ID context)

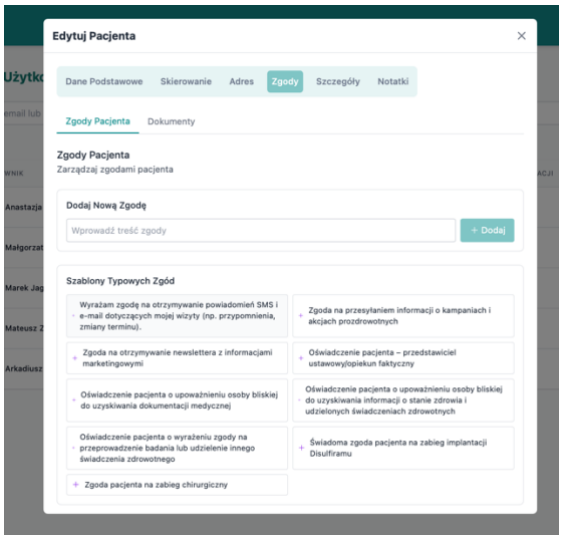
- **Dane podstawowe**
- **Zgody**

Applies to:

- scheduled visits,
- cancelled visits,
- no-show visits,
- online and reception registrations.

These tabs must NEVER be hidden. (we need to have access to them in the Visite ID, patient ID, as well as canceled visite ID data, etc.)





5. PESEL VALIDATION – INCORRECT "FIRST VISIT" REGISTRATION SCENARIO

Situation Description

A situation may occur where:

- the reception staff mistakenly registers a patient as a **"First visit"**,
- the patient already exists in the system (has a **PATIENT_ID**),
- the PESEL number is entered only later, at the **"Edit patient"** stage.

This is an operational mistake by the reception staff that the system must handle **safely and automatically**, without creating duplicate patients.

Required System Behavior

Step 1 – Entering the PESEL number

- After entering a **complete PESEL number (11 digits)**, the system:
    - checks whether a patient with this PESEL already exists in the database.

Step 2 – Existing patient detected

If a patient with the given PESEL is found, the system:

- does **NOT** create a new **PATIENT_ID**,
- does **NOT** automatically overwrite any data,
- displays an informational message (not an error):

„Pacjent o podanym numerze PESEL już istnieje w systemie."

- displays an action button:

[Załaduj dane istniejącego pacjenta]

Step 3 – Reception staff action

- Only after the reception staff **explicitly clicks the button**:
    - the form is reloaded with data from the existing **PATIENT_ID**,
    - previously entered form data (e.g. name, surname) may be overwritten,
    - PESEL and identity fields may be locked for editing (optional).

Step 4 – Completing registration

- After clicking **"Zakończ rejestrację"**:
  - the system assigns **VISIT_ID → existing PATIENT_ID**,
  - the system does **NOT** create a new patient,
  - any patient data changes are saved.

Overriding Rules (Important)

- **PESEL always takes precedence** over the selected "First / Follow-up visit" flow.
- The selected UI flow **does not determine patient identity**.
- This validation:
  - acts only as a safety mechanism (edge case),
  - is **not** a normal workflow scenario,
  - prevents the creation of duplicate patient records.

What the System MUST NOT Do

- ❌ create a second patient account
- ❌ require manual record merging
- ❌ interrupt the process with an error
- ❌ overwrite data without explicit user action

Summary (One Sentence)

If the reception registers a visit as "first", but after entering PESEL it turns out that the patient already exists, the system informs the user, allows manual loading of the existing patient data, and after completing registration assigns the VISIT_ID to the existing PATIENT_ID without creating duplicates.

PESEL Validation

The **PESEL** field must be subject to basic technical validation to detect potential data entry errors.

Rules:

- the PESEL field must accept **digits only (0–9)**,
- the input length must be **strictly limited to 11 characters**,

- after entering **11 digits**, it must **not be possible to enter additional characters**.

Additionally, the system performs **PESEL checksum validation using the official algorithm**.

If the PESEL **does not pass the checksum validation**:

- the system must display a **warning message**:

  "Warning: the PESEL number may be invalid."

- patient registration **must NOT be blocked**,
- the user must be allowed to **continue the registration process intentionally**.

PESEL validation:

- is **informational only (soft validation)**,
- is used solely to detect possible typing errors,
- **must not prevent patient registration completion**,
- does not rely on any external or governmental databases.

6. PATIENT_ID is created ONLY after clicking "Complete registration".

After clicking **"Complete registration"**:

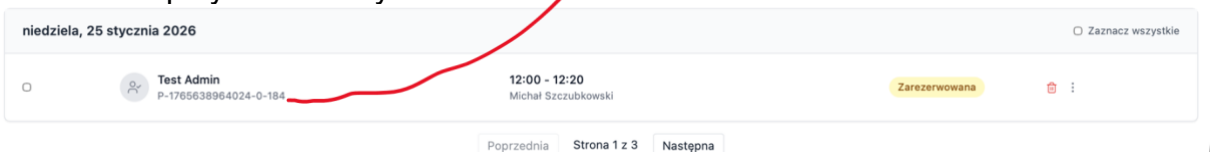
- if the **PESEL is new**:
  - create a new patient record,
  - create a **PATIENT_ID**.
- if the **PESEL already exists**:
  - do **NOT** create a new patient,
  - assign:
    VISIT_ID → existing PATIENT_ID.

8. "UNVERIFIED" STATUS

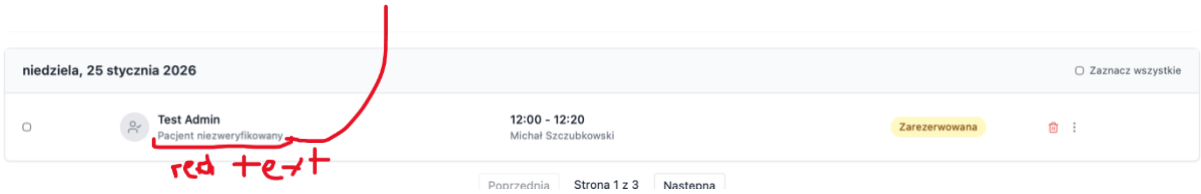
UX behavior:

- In the place where **PATIENT_ID** is normally displayed:
  - if **PATIENT_ID does not exist**, display:
    "Pacjent niezweryfikowany".
- After **PATIENT_ID is created**:
  - display the standard **Patient ID view** (read-only).

- the view of a patient who has a patient ID and a visit ID assigned to the patient ID, the ID is displayed normally as it is now



- a patient who does not yet have a patient ID (no PESEL number entered) and only has the status of visite ID without assigning a visit to a patient ID



9. CANCELLATION / NO-SHOW

If the patient never showed up:

- no PESEL,
- no PATIENT_ID,
- no USERNAME.

System behavior:

- the visit remains as a **VISIT_ID**,
- status: **CANCELLED / NO-SHOW**,
- the following remain accessible:
  - Basic data,
  - Consents.

Ghost / unused patient accounts must NOT be created.

10. PATIENT ACCOUNT – LOGIN (TO BE ENABLED LATER)

Rule

A patient account **cannot be created** if the patient has never visited the clinic.

Login flow

1. The patient opens the Patient Portal login page.
2. The patient clicks **Log in** or **Create account**, and the system requests data required to create an account.

3. The patient enters **PESEL**.
4. The system checks:
    - whether a **PATIENT_ID exists** (i.e. whether the patient exists in the system and has visited at least once).
5. If **YES**:
    - the patient enters an email address to be associated with the patient account,
    - the system sends login details or a login link to this email, including a temporary password,
    - the temporary password can later be changed, just like the username,
    - if the patient changes the username in their account, the updated username must also be reflected in the internal system/dashboard.
6. If **NO** (the patient has no PATIENT_ID and therefore no medical patient record):
    - display the message:
      "No patient account found – please contact the reception desk."

→ No registration of users who have never been patients.

11. PATIENT LIST – RULES

Important:
Currently, the patient list includes records derived from all visit statuses. This behavior must be changed.

Core rule:

- **Only patients who have a PATIENT_ID must appear in the patient list.**

This means that:

- visits without a created PATIENT_ID (first visit, cancelled, no-show),
- persons who have never physically visited the clinic,

MUST NOT appear in the patient list.

The patient list must display only:

- patient full name,
- **PATIENT_ID**,
- date and time of the **first completed visit**,
- the physician assigned to the first visit.

Empty / placeholder patient accounts must NOT be created, because:

- they do not represent real patients,
- they distort statistics,
- they make it impossible to reliably determine the total number of patients in the system.